

NAME: N. Kalyan Reddy

REG NO: 192424232

SLOT: C

Intelligent AI-Based Smart Parking and Vehicle Guidance System

1. Understanding the Problem Statement

1.1 Problem Explanation

Nowadays, the number of vehicles in cities is increasing rapidly because of urbanization. One of the biggest problems faced by drivers is finding an empty parking space. Many people spend several minutes driving around parking areas searching for a vacant slot. This not only wastes time but also increases traffic congestion, fuel consumption, and air pollution.

Most traditional parking systems do not provide real-time information about available parking spaces. As a result, drivers often enter full parking lots without knowing whether any space is available.

The proposed Intelligent AI-Based Smart Parking and Vehicle Guidance System aims to solve this problem by using Artificial Intelligence (AI), IoT sensors, cameras, GPS, cloud computing, and a mobile application. The system can automatically detect vacant parking spaces, guide drivers to available slots, predict parking demand, allow online reservations, and provide digital payment facilities.

1.2 Objectives of the Proposed System

The main objectives of the proposed system are:

- Detect available parking spaces automatically.
- Guide drivers to the nearest vacant parking slot.
- Predict parking demand using AI.
- Allow users to reserve parking spaces in advance.
- Support secure digital payment methods.
- Monitor parking occupancy in real time.
- Generate parking usage reports for administrators.

1.3 Scope of the Problem

The proposed system can be used in places where parking management is difficult, such as:

- Shopping malls
- Hospitals
- Airports
- Railway stations
- Universities
- Office buildings
- Smart cities

The system mainly focuses on parking management, reservation, payment, navigation, and occupancy monitoring. It does not include vehicle servicing, insurance management, or traffic law enforcement.

2. Stakeholder and Challenge Analysis

2.1 Stakeholders

The major stakeholders involved in this system are:

- Drivers
- Parking Operators
- System Administrator
- City Administration
- Payment Service Provider
- AI System
- IoT Sensors and Cameras
- Maintenance Team

2.2 Roles and Responsibilities

Drivers

Drivers use the mobile application to check parking availability, reserve parking spaces, navigate to the location using GPS, make online payments, and view their parking history.

Parking Operators

Parking operators manage the parking area, monitor occupied and vacant spaces, and ensure smooth parking operations.

System Administrator

The administrator manages users, parking locations, system settings, reports, and monitors the overall performance of the application.

City Administration

The city authorities analyze parking utilization reports and traffic data to improve urban planning and parking management.

AI System

The AI module analyzes parking occupancy, predicts future parking demand, and recommends suitable parking spaces.

IoT Sensors and Cameras

Sensors and cameras detect whether a parking slot is occupied or vacant and update the information in real time.

Payment Gateway

The payment gateway securely processes online transactions through UPI, debit cards, credit cards, or digital wallets.

Maintenance Team

The maintenance team is responsible for repairing sensors, cameras, and other hardware components whenever necessary.

2.3 Challenges Faced by Stakeholders

Drivers often waste time searching for parking spaces, leading to frustration and extra fuel costs.

Parking operators find it difficult to monitor parking manually, especially during busy hours.

City administrators face challenges in reducing traffic congestion and improving parking utilization.

System administrators must ensure that the system runs smoothly without technical issues.

The AI system depends on accurate data for making reliable predictions.

IoT devices require regular maintenance because damaged sensors may provide incorrect parking information.

3. Current System Analysis

3.1 Existing System

Most existing parking systems are managed manually. Drivers enter the parking area and search for available spaces by themselves. Parking attendants may guide vehicles, but there is no guarantee that information about vacant spaces is accurate.

Payments are often collected manually or through ticket machines, and there is usually no facility for online booking or reservation.

3.2 Strengths of the Existing System

The existing system is simple and easy to operate. It does not require expensive technology and works reasonably well in small parking areas with fewer vehicles.

3.3 Limitations

However, the traditional system has many disadvantages.

- Drivers spend too much time searching for parking.
- Traffic congestion increases near parking areas.
- Fuel is wasted unnecessarily.
- There is no real-time parking information.
- Online reservation is not available.
- Manual monitoring may lead to errors.
- There is no demand prediction or occupancy analysis.

3.4 Problems Identified

After analyzing the current system, the following problems were identified:

- Lack of automation
- No intelligent parking guidance
- Poor utilization of parking spaces
- Long waiting time during peak hours
- No online reservation
- Limited reporting facilities
- Increased operational cost

4. Proposed Software Solution

4.1 Proposed Solution

The proposed system uses AI and IoT technologies to provide a smart parking solution. Cameras and sensors continuously monitor parking spaces and identify vacant slots. The information is updated instantly in a cloud database.

Users can open the mobile application to view available parking spaces, reserve a slot, navigate using GPS, and complete online payment before arriving.

The AI module analyzes historical parking data to predict parking demand and improve parking allocation during busy periods.

4.2 Functional Modules

The proposed system consists of the following modules:

User Management

- Registration
- Login
- Profile Management

Parking Detection Module

- AI camera detection
- IoT sensor monitoring
- Real-time parking updates

Reservation Module

- Book parking
- Cancel reservation
- View booking history

Navigation Module

- GPS guidance
- Nearest parking suggestion

Payment Module

- Online payment
- Payment confirmation
- Transaction history

AI Prediction Module

- Parking demand prediction
- Peak-hour analysis

Administrator Module

- User management
- Parking management
- Report generation

Analytics Module

- Parking utilization reports
- Revenue reports
- Occupancy analysis

4.3 Functional Requirements

The system should allow users to:

- Register and log in.
- View available parking spaces.
- Reserve parking slots.
- Cancel reservations.
- Make digital payments.
- Receive parking notifications.
- Navigate using GPS.
- View booking history.
- Generate occupancy reports.

4.4 Non-Functional Requirements

The proposed system should provide:

- Fast response time
- High reliability
- Strong security
- Easy maintenance
- User-friendly interface
- Scalability for multiple parking locations
- High system availability

4.5 Software Engineering Principles

The proposed system follows important software engineering principles.

Modularity: Different functions such as reservation, payment, navigation, and AI prediction are developed as separate modules.

Scalability: The system can support additional parking locations without major modifications.

Maintainability: Modular design makes future updates and maintenance easier.

Reliability: Real-time monitoring ensures accurate parking information.

Usability: The mobile application is designed with a simple and easy-to-use interface.

Security: User information and payment details are protected using secure authentication and encryption.

5. Justification of SDLC Model

Selected Model: Agile Scrum

Agile Scrum is the most suitable software development methodology for this project because smart parking systems require continuous improvements based on user feedback and technological advancements.

The project can be developed in small iterations called sprints. Each sprint focuses on implementing one major feature, such as user management, parking detection, reservation, payment, or AI prediction.

This approach allows developers to test features early, identify issues quickly, and make improvements without affecting the entire system.

Agile also encourages teamwork, continuous testing, and regular customer feedback, which increases the chances of delivering a successful software product.

6. Expected Outcomes and Limitations

6.1 Expected Outcomes

The proposed system is expected to provide the following benefits:

- Reduced parking search time
- Less traffic congestion
- Lower fuel consumption

- Reduced carbon emissions
- Better parking space utilization
- Improved user experience
- Increased parking revenue
- Better decision-making through analytics

6.2 How the Solution Solves the Problem

The AI-based parking detection system provides real-time information about available parking spaces, helping drivers find parking quickly.

GPS navigation directs users to the nearest available parking location, reducing unnecessary driving.

Online reservation eliminates uncertainty, while digital payments make the parking process faster and more convenient.

The analytics dashboard helps administrators monitor parking usage and improve overall parking management.

6.3 Limitations

Although the proposed system offers many advantages, it also has some limitations.

The installation cost of AI cameras and IoT sensors is relatively high. The system also depends on a stable internet connection. Hardware failures or inaccurate sensor readings may affect system performance. In addition, regular maintenance is required to ensure reliable operation.

6.4 Future Enhancements

The system can be further improved by adding:

- Electric vehicle charging station support
- Automatic number plate recognition
- Voice-based parking assistance
- Dynamic parking pricing
- Integration with autonomous vehicles
- Smart city traffic management systems
- AI-based traffic prediction
- Multi-city parking support

Conclusion

The Intelligent AI-Based Smart Parking and Vehicle Guidance System is an effective solution for addressing modern parking challenges. By combining Artificial Intelligence, IoT, GPS, cloud computing, and digital payment technologies, the system provides real-time parking information, reduces traffic congestion, improves parking utilization, and enhances the overall user experience.

The Agile Scrum development methodology makes the project flexible and easy to improve over time. Overall, this system supports the vision of smart cities by making parking management more efficient, reliable, and environmentally friendly.